



Amirhossein Rezaei

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EDUCATION

University of Padova

Physics of Data (M.Sc.)

Padova, Italy

Oct 2025 – Present

National University of Iran (Shahid Beheshti University)

B.Sc. in Physics; GPA: 3.54/4.00

Tehran, Iran

Sep 2019 – Feb 2024

Shahid Madani High School, NODET [↗](#)

National Organization for Development of Exceptional Talents [↗](#)

Tabriz, Iran

Sep 2013 – Jun 2017

RESEARCH EXPERIENCE

Research Papers

- A 1D-CNN deep learning framework for seismic collapse prediction of jacket offshore platforms with Bayesian neural architecture search. *Ocean Engineering* 352, 124625. [↗](#)
- Continuous Approximation of the Ising Hamiltonian: Exact Ground States and Applications to Fidelity Assessment in Ising Machines. *Physical Review E*, 112(2), 024102. [↗](#)
An intuitive and very accessible video explanation of the paper's key concepts and results is available on [YouTube](#): [Watch here](#) [↗](#)
- The generation of baryon asymmetry and hypermagnetic field by the chiral vortical effect in the presence of sphalerons. *Nuclear Physics B* (2025): 116895. [↗](#)
- Effects of Nonhelical Component of Hypermagnetic Field on the Evolution of the Matter-Antimatter Asymmetry, Vorticity, and Hypermagnetic Field. *Physical Review D* 104 (5). [↗](#)

Teaching Assistant

- Physics I, Assistant to Dr. Sadeghi

RELEVANT COURSES AND GRADES

Quantum Information and Computations	16.8/20
Quantum Mechanics III	18.51/20
Introduction to Artificial Intelligence	17.5/20
Biophysics	20/20
Complex Systems	19/20
Mathematical Physics II	19.5/20
Physics Project	20/20
Foundations of Numerical Simulations	20/20
Nuclear and Elementary Particle Physics	17.51/20
Thermodynamics and Statistical Mechanics I	17.3/20
Analytical Mechanics I	17.75/20
General Topology	14.5/20

For the full grades, see [here](#) [↗](#).

SKILLS

Programming: Python, C, C++, Fortran, MATLAB, Mathematica, Maple

Tools and Technologies: PyTorch, TensorFlow, Keras, Scikit, SciPy, Pandas, MNE, Matplotlib, NumPy, Git, L^AT_EX

Languages:

- Persian (Native)
- English (C1 – IELTS Overall Band 7.5; Writing: 7.5, Listening: 7.5, Reading: 8, Speaking: 7)
- Azerbaijani
- Turkish

PROJECTS

Sagittarius: Cinematic Black Hole Renderer | [GitHub](#)

- Authored a ground-up rewrite of the Gargantua renderer, migrating the backend from Taichi to NVIDIA Warp to achieve direct CUDA control and a highly optimized, multi-kernel architecture.
- Upgraded the 3D fluid simulation by implementing a BFECC (Back and Forth Error Compensation and Correction) advection scheme, drastically reducing numerical diffusion and preserving sharp gas filaments.
- Optimized ray-marching performance by pre-baking high-frequency FBM noise fields directly into VRAM, replacing costly per-pixel evaluations with single trilinear grid samples.
- Enhanced physical accuracy and lighting by calculating relativistic Doppler beaming and blackbody radiation temperature shifts against the bent photon geodesic direction.

Gargantua: Cinematic Black Hole Renderer | [GitHub](#)

- Developed a real-time, cinematic black hole renderer in Python, leveraging the Taichi language to JIT-compile high-performance CUDA kernels for the GPU.
- Engineered a physics-based backend featuring a general relativistic ray tracer (using a Dormand-Prince 5(4) adaptive integrator) and a custom 3D Eulerian fluid simulation on a cylindrical coordinate grid for the accretion disk.
- Integrated advanced volumetric rendering, a full cinematic post-processing suite (ACES tone mapping, bloom, anamorphic flares), and an intelligent frame caching system to optimize workflow and visual fidelity.

SoME4 Video Project | [GitHub](#)

- An animated explainer video for the [2025 Summer of Math Exposition \(SoME4\)](#), built with [Manim](#) and [PsiCreature](#), showcasing research in Ising machine fidelity through storytelling and custom character animation.
- The video explains the paper “*Continuous Approximation of the Ising Hamiltonian: Exact Ground States and Applications to Fidelity Assessment in Ising Machines*”.
- The YouTube video is available [here](#)

Divar Scraper with Multithreading | [GitHub](#)

- Divar Scraper with Multithreading: This project allows you to scrape data from the Divar website, specifically for real estate listings in different categories and cities. It utilizes the Divar API to retrieve and store information about real estate listings in an Excel file. This module provides a command-line tool for scraping posts from a website.

TGJU-API | [GitHub](#)

- TGJU-API: A Python web scraper for retrieving currency and gold prices from the tgju.org website.

Complex-Systems | [GitHub](#)

- A repository of Complex Systems ideas and codes.

Optimizing the Ising model in Python | [Medium](#)

- This project involves simulating a 2D ISING model using various techniques such as utilizing the fast Numba JIT Compiler, implementing parallel processing with Numba's prange, performing branch-less computation, utilizing a lookup table for the exponential function, and optimizing further with a custom random number generator.

RELATED ACTIVITIES

Physics Student Association of National University of Iran (2022-2023)

- Elected Member, Main Council of Physics Student Association, National University of Iran
- Volunteered and campaigned for the position, receiving support from students through votes.
- Coordinated events and activities for students interested in physics and related fields.
- Organized weekly seminars and workshops featuring guest speakers from academia and industry.
- Collaborated with other student organizations to promote science education and outreach in the community.

Representative of the Faculty of Physics and Adviser to the University President (2023-2024)

- Selected Member, Representative of the Faculty of Physics, acting as a voice for the student body.
- Actively involved in advocating for student demands and interests within the faculty.
- Played a key role in enhancing the welfare and academic environment of the Physics department.
- Organized and coordinated events, seminars, and workshops for students and faculty.